

HealthConnect Clinic, Week 8

Week 7 → Week 8 Transition (Data Analytics Track)

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1. Main Week 7 output

The Analytics Testing & Refinement Report: all five KPIs independently recomputed and validated, the lead time and distance findings retested against new segments, a cross track test of the DS track collaborator's `double_risk_flag` feature, and a design (not a run) of a double risk call pilot.

2. Most important testing result

All five dashboard KPIs were recomputed directly from the raw prepared dataset, bypassing the notebook entirely, and matched to four decimal places. This confirmed the pipeline itself is correct, not an artefact of one notebook.

3. Major issue discovered

The double risk segment did not reproduce on first attempt: coding it as lead time ≥ 46 days gave 2.46% of bookings at 82.1% risk, against the reported 4.66% at 73.0%. The root cause was the same: the DS track collaborator's `double_risk_flag` feature had also been built on the ≥ 46 threshold.

4. What was refined

The double risk definition was corrected to ≥ 31 days and retested under both a Random Forest and the exact Week 6 Logistic Regression candidate, confirming no performance difference between the two thresholds. The 31+ definition was kept because it reaches nearly twice as many high risk patients (233 vs 123 at ≥ 46 days) at a still very high no-show rate (73.0% vs 82.1%). The 15km+ distance effect was confirmed as real and independent of lead time, though modest. The KPI 5 (cancellation to no-show ratio) testing gap was also closed.

5. Component now ready

The four KPI charts with confidence interval error bars, the KPI 5 ratio chart, the lead time and distance findings, and the corrected double risk segment (233 patients, 73.0% no-show rate, bootstrap CI [67.4%, 79.0%]) are tested, validated, and ready for final integration.

6. What still needs to be integrated

The distance chart's bar colour refinement was one of the expected Week 8 outputs, but applying it was decided against: it is a purely visual styling change (colour grouping and an annotation) that does not alter any KPI, finding, or result, so it was dropped as not worth the effort for a change with no material effect on the output. The DS track collaborator still needs to merge the corrected ≥ 31 threshold into `train_improved.py`.

7. Track(s) involved in final integration

Data Analytics (primary) and the DS track collaborator, for the `double_risk_flag` correction.

8. What I intend to demonstrate

How testing caught and diagnosed a real discrepancy in the double risk definition, how the fix was validated against two different model types with no loss in performance, and the final validated KPI and dashboard package as the decision support handoff to the DS track.